FINANCIAL® DATA EXCHANGE

CSDF Tagging Guide

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Contents

1 INTRODUCTION 4

1.1.1 Reference Lists for FDX Consensus Standard Data Format (CSDF) 4
1.1.2 FDX adoption of Consensus Standard Data Format Reference List 4
1.1.3 Reference Lists in the FDX Consensus Standard Data Format Guidelines 5
1.2 THE FDX CUSTOM EXTENSIONS 5
1.2.1 Specification extensions as tags 6
1.2.2 Extension full definitions 7
1.2.3 Tagged files 8
1.3 FDX EXTENSION EXAMPLES 9
1.3.1 AccountDescriptor [ANY] tags 9
1.3.2 DepositAccount [REG_E] and LineOfCreditAccount [REG_Z] tags 11
1.3.3 PaginatedArray technical tags for non-Account optional elements 12
1.3.4 Availability technical tags for non-Account data 12
1.3.5 Path and operation tags 13
1.3.6 Identifying needed but untagged elements 14
1.4 CSDF IMPLEMENTATION USING ONLY TAGGED ELEMENTS 15
1.5 RELATED RESOURCES 16

1 Introduction

Financial Data Exchange (FDX) defines a set of APIs as a common standard for the secure and convenient access of permissioned consumer and business financial data.

FDX API release V6.4 defines a subset of the FDX APIs called the **Consensus Standard Data Format (CSDF)**. The CSDF is laid out in the FDX Consensus Standard Data Format Guidelines and defines specific requirements for Data Providers' developer interfaces to conform to standards adopted under the Consumer Financial Protection Bureau's (CFPB) Personal Financial Data Rights Rulemaking.

The FDX CSDF contains subsets of the FDX Core, Customer, Meta and Money-Movement APIs. FDX has used custom FDX-defined OpenAPI Specification extensions to capture these subsets directly in the V6.4 and V5.4 files of the FDX API Spring Release 2025. This CSDF Tagging Guide describes how Data Providers can interpret and use these extensions to help ensure delivery of customer data in a way that aligns with CSDF requirements.

> **Note**: The CSDF Tagging strategy has not changed for the latest release, now FDX API v6.5 – Fall 2025 Release. While API v6.4 is referenced throughout this document, all information also applies to API v6.5 and Reference List v6.5.

1.1.1 Reference Lists for FDX Consensus Standard Data Format (CSDF)

The FDX API v6.4 - Spring 2025 Release defines a Reference List identifying specific schemas and properties of the FDX APIs which are candidates for the CSDF to meet the Consumer Financial Protection Bureau (CFPB) regulations for Section 1033 of the Dodd-Frank Act.

The simultaneous release of FDX API v5.4 also defines its own closely related Reference List.

1.1.2 FDX adoption of Consensus Standard Data Format Reference List

Summary

This introduces the **Reference List** for the Consensus Standard Data Format (CSDF) laid out in the FDX Consensus Standard Data Format Guidelines document.

Problem Statement

Data providers lack clear guidance in the FDX API on which data elements are relevant to CFPB regulations for Section 1033 of the Dodd-Frank Act.

- Regulation E accounts, referred to as Reg E, are deposit accounts such as savings, checking and others which are covered by regulations described in the Electronic Fund Transfer Act
- Regulation Z accounts, referred to as Reg Z, are loan accounts as covered by regulations described in the Truth In Lending Act

Impact/ Benefits of implementing the Reference List

The reference list offers a comprehensive compilation of data elements from the FDX API that are applicable and available for Regulation E and/or Regulation Z accounts.

Details

The created V6.4 and V5.4 Reference List spreadsheets designate the relevant FDX specification data elements as TRUE/FALSE for coverage by Regulation E and/or Regulation Z accounts or for Technical use* in meeting CFPB regulations. The Reference Lists are a resource for Data Provider implementation teams, helping them identify which fields may be relevant for their APIs.

* *Technical* use tag is defined in next section

1.1.3 Reference Lists in the FDX Consensus Standard Data Format Guidelines

For clarity, a Data Provider is not obligated to return all of the data elements in the Reference List.

- A Data Provider may share the subset of the Reference List that the Data Provider deems to that it is obligated to share under the PFDR Rulemaking.*
- Alternatively, a Data Provider may share all of the Reference List elements, as well as other elements defined in the full FDX API spec.

* FDX has not sought to define a consensus view on which specific data elements are "covered data" under the PFDR Rulemaking. Because the PFDR Rulemaking at times defines "covered data" in high-level terms, different parties may reasonably reach different interpretations of whether a specific data element is "covered data" that must be shared.

1.2 The FDX Custom Extensions

The FDX API adds a new `fdxapi.extensions.yaml` file which follows the OpenAPI Specification but only documents the new FDX specification extensions used to designate the CSDF subset of the FDX API. The file is not itself part of the FDX API.

The FDX specification extensions capture the content of the Reference Lists and translates them into the FDX API specification.

1.2.1 Specification extensions as *tags*

FDX has added specification extensions to the Reference List elements in the FDX OpenAPI Specification: schemas, properties, paths and operations. In the context of each element's definition, the specification extension denotes that element as a member of the Reference List. For convenience we call that extension denotation of the given element a *tag*. Files and schema elements containing a tag are said to be *tagged*. These tags indicate whether those contextual OpenAPI elements of the FDX API could be included in a Data Provider's implementation of a CSDF API.

Extension name and value	Reference List mapping
The ***Account Categories*** tag takes an array value: `x-fdx-csdf-account-categories: [...]`	Tags a path, operation, schema or property as on the Reference List of the given array of account types. For initial V6.4 and V5.4 versions, only a single value is needed for each tag, as shown in following rows
`x-fdx-csdf-account-categories: [REG_E]`	Tags this path, operation, schema or property as on the Reference List of Regulation E accounts (e.g. deposit accounts)
`x-fdx-csdf-account-categories: [REG_Z]`	Tags as on the Reference List of Regulation Z accounts (e.g. credit card accounts)
`x-fdx-csdf-account-categories: [ANY]`	Tags as on the Reference List of both Regulation E and Regulation Z accounts, including Digital Wallets which can contain either or both
The ***Technical*** tag takes a boolean value: `x-fdx-csdf-technical: true`	Tags either: • An optional, technical field which supports delivery of Reg E or Reg Z account data such as pagination elements, OR • A field which supports delivery of non-account-related data from the Reference List, such as Availability metrics

`false` is “invalid” and never used: `x-fdx-csdf-technical: false`	Not needed. Instead, the defined element is not tagged at all, and absence of a tag indicates an element is not on the Reference List.
Also valid as mentioned above is `x-fdx-csdf-account-categories: [REG_E, REG_Z]`, not used in V6.4 or V5.4 releases because it is equivalent to `x-fdx-csdf-account-categories: [ANY]`.	

1.2.2 Extension full definitions

The new `fdxapi.extensions.yaml` file as delivered in these releases V6.4 and V5.4, see following content for explanations and discussion:

```
openapi: 3.1.0
info:
  version: '6.4.0'
  title: FDX V6.4.0 API Extensions
  description: Defined extensions to FDX V6.4.0 API
  contact:
    name: Financial Data Exchange
    url: 'https://financialdataexchange.org/'
    email: fdxsupport@financialdataexchange.org
  license:
    name: FDX API License Agreement (11-13-2019)
    url: 'https://financialdataexchange.org/common/Uploaded%20files
          /Policies/FDX%20API%20License%20Agreement-(11-13-2019).pdf'
  x-fdx-csdf-technical: true
# servers: {}

# paths: {}

components:

  schemas:
    ##############################################################
    #
    # Defined schemas for FDX extensions
    #
    ##############################################################

    CsdfAccountCategory:
      title: Account Category enum for FDX CSDF extension tags
      description: The values for the x-fdx-csdf-account-categories
        FDX extension tag to mark all the FDX API entities which are
        candidates for U.S. CFPB section 1033 Personal Financial Data
        Rules (PFDR) regulations for Consensus Standard Data Format
        (CSDF), according to Data Provider's interpretation.
        One or more of `ANY`, `REG_E` or `REG_Z`.
          * REG_E - The deposit accounts and funds transfer
```

```
            activities to which U.S. banking Regulation E
            "Electronic Funds Transfer Act" (EFTA) applies
          * REG_Z - The loan and line of credit accounts and
            credit transaction activities to which U.S. banking
            Regulation Z "Truth in Lending Act" (TILA) applies
          * ANY - The FDX-defined entity and activity data to which
            any of these mentioned U.S. banking regulations apply
      type: string
      enum:
        - ANY
        - REG_E
        - REG_Z

    x-fdx-csdf-account-categories:
      title: FDX CSDF Account extension tag
      description: The FDX x-fdx-csdf-account-categories extension
        tag marks all the FDX API entities and properties which are
        candidates for U.S. CFPB section 1033 Personal Financial
        Data Rules (PFDR) regulations for Consensus Standard Data
        Format (CSDF). One or more of `ANY`, `REG_E` or `REG_Z`,
        as defined by `CsdfAccountCategory`
      type: array
      items:
        $ref: '#/components/schemas/CsdfAccountCategory'

    x-fdx-csdf-technical:
      title: FDX CSDF Technical extension tag
      description: The FDX x-fdx-csdf-technical extension tag marks
        all the FDX API entities and properties which may be needed
        for technical implementation reasons to deliver the entities
        and properties which are on the FDX Reference List for U.S.
        CFPB section 1033 Personal Financial Data Rules (PFDR)
        regulations for a Consensus Standard Data Format (CSDF).
        If `true` indicates that the entity or property likely should
        be included by a Data Provider
      type: boolean
```

1.2.3 Tagged files

The tagging of the FDX APIs begins with the five affected API files, plus the extension definition file shown above:

File	Tag
fdxapi.extensions.yaml	x-fdx-csdf-technical: true
fdxapi.components.yaml	x-fdx-csdf-account-categories: [ANY]

fdxapi.core.yaml	x-fdx-csdf-account-categories: [ANY]
fdxapi.customer.yaml	x-fdx-csdf-account-categories: [ANY]
fdxapi.meta.yaml	x-fdx-csdf-technical: true
fdxapi.money-movement.yaml	x-fdx-csdf-account-categories: [REG_E]

1.3 FDX extension examples

Following see specific examples of extensions from the V6.4 release.

1.3.1 AccountDescriptor [ANY] tags

From the `fdxapi.core.yaml` file, the base `AccountDescriptor` schema and its properties tagged with `[ANY]` apply to all of the regulated account types (i.e. Reg E and Reg Z) which are composed from it and are then themselves tagged. We've used `[ANY]` here so that future account type peers of Reg E and Reg Z will automatically be included. (Some source lines removed for simplicity, and the tag comments are added here only.)

```
components:
  schemas:

    AccountDescriptor:
      title: Account Descriptor entity
      description: This descriptor provides minimal information about
        the account for use in lightweight arrays
      type: object
      x-fdx-csdf-account-categories: [ANY]    # <-- Schema tag "ANY"
      properties:
        accountCategory:
          $ref: '#/components/schemas/AccountCategory'
          description: The entity type of this Account
          x-fdx-csdf-account-categories: [ANY]
                                          # ^^^ Property tag "ANY"
        accountId:
          $ref: '#/components/schemas/Identifier'
          description: >-
            Long-term persistent identity of the account, though not
            an account number.
            This identity must be unique to the owning institution
          x-fdx-csdf-account-categories: [ANY]
                                  # Another ^^^ property tag "ANY"
# ...
```

In this example, `AccountDescriptor` is a “schema definition” and then we would say it is a “tagged schema”. Both `accountCategory` and `accountId` are “property definitions”, and then “tagged properties”.

1.3.2 DepositAccount [REG_E] and LineOfCreditAccount [REG_Z] tags

The specific `DepositAccount` and `LineOfCreditAccount` schemas separately apply to just one of the regulated account types. Both also inherit properties from `Account` schema, which like `AccountDescriptor` is tagged with `[ANY]`. (Some source lines removed for simplicity.)

```
    DepositAccount:
      title: Deposit Account entity
      description: Information for a deposit account type
      type: object
      x-fdx-csdf-account-categories: [REG_E] # <-- Schema tag "REG_E"
      allOf:
        - $ref: '#/components/schemas/Account'
        - type: object
          properties:
            balanceAsOf:
              description: As-of date of balances
              $ref: '#/components/schemas/Timestamp'
              x-fdx-csdf-account-categories: [REG_E]
                                             # ^^^ Property tag "REG_E"
# ...

    LineOfCreditAccount:
      title: Line of Credit Account entity
      description: A line of credit account
      type: object
      x-fdx-csdf-account-categories: [REG_Z] # <-- Schema tag "REG_Z"
      allOf:
        - $ref: '#/components/schemas/Account'
        - type: object
          properties:
            bills:
              type: array
              description: Payments due on the account
              x-fdx-csdf-account-categories: [REG_Z]
                                             # ^^^ Property tag "REG_Z"
              items:
                $ref: '#/components/schemas/Bills'
# ...
```

If Data Provider A holds only Deposit accounts, and Data Provider B holds only Line of Credit accounts, and Data Provider C holds both Deposit accounts and Line of Credit accounts, then each of the three providers would need to consider all elements tagged `[ANY]`. Data Provider A would not need to implement any elements tagged `[REG_Z]` and Data Provider B would not need to implement any elements tagged `[REG_E]`. When implementing a CSDF none of the Data Providers would need to implement elements that are not tagged at all, other than those they find when Identifying needed but untagged elements as discussed below.

1.3.3 PaginatedArray technical tags for non-Account optional elements

From the `fdxapi.components.yaml` file, the `PaginatedArray` schema and properties can be used to return paginated sets of results from an API operation. As such, they have the "technical" tag. This indicates they can optionally be used to support Reg E or Reg Z account data, but ***do not themselves represent account data***.

```
    PaginatedArray:
      title: Paginated Array
      description: Base class for results that may be paginated
      type: object
      x-fdx-csdf-technical: true    # <-- Schema technical tag
      properties:
        page:
          $ref: '#/components/schemas/PageMetadata'
          description: Offset IDs for navigating result sets
          x-fdx-csdf-technical: true    # <-- Property technical tag
        links:
          $ref: '#/components/schemas/PageMetadataLinks'
          description: Resource URLs for navigating result sets
          x-fdx-csdf-technical: true
                        # ^^^ Another property technical tag
```

1.3.4 Availability technical tags for non-Account data

From the `fdxapi.meta.yaml` file, the `AvailabilityList` and `Availability` schemas and properties and the `/availability` path are used to return non-Account data about the provider's API availability and outages. These have the other use of the "technical" tag to indicate they can be used to support non-Account data.

```
    AvailabilityList:
      title: Availability List entity
      description: Response object for /availability API
      type: object
      x-fdx-csdf-technical: true    # <-- Schema technical tag
      properties:
        availability:
          description: Zero or more API availability metrics
          type: array
          x-fdx-csdf-technical: true    # <-- Property technical tag
          items:
            $ref: '#/components/schemas/Availability'
```

1.3.5 Path and operation tags

Back to the `fdxapi.core.yaml` file, the `/accounts/{accountId}/payment-networks` path and `getAccountPaymentNetworks` operation are tagged `[REG_E]` to indicate they are required when returning deposit account data. (Assuming the provider has determined that the operation's response data elements do represent Covered Data under section 1033.)

The `/availability` path in the `fdxapi.meta.yaml` file looks similar (not shown here), but it uses the `x-fdx-csdf-technical: true` tag.

```
  /accounts/{accountId}/payment-networks:
    parameters:
      - $ref: ' #/components/parameters/FapiInteractionIdHeader'
      - $ref: ' #/components/parameters/FdxApiActorTypeHeader'
      - $ref: ' #/components/parameters/FdxApiDataProviderIdHeader'
    x-fdx-csdf-account-categories: [REG_E]    # <-- Path tag "REG_E"
    get:
      operationId: getAccountPaymentNetworks
      tags:
        - Money Movement
      description: Get payment networks supported by the account
      summary: Get payment networks supported by the account
      x-fdx-csdf-account-categories: [REG_E]
                                   # ^^^ Operation tag "REG_E"
      parameters:
        - $ref: '#/components/parameters/AccountIdPath'
        - $ref: '#/components/parameters/OffsetQuery'
        - $ref: '#/components/parameters/PageKeyQuery'
        - $ref: '#/components/parameters/LimitQuery'
      responses:
        '200':
          description: Information required to execute a payment
            transaction against this account
          headers:
            x-fapi-interaction-id:
              $ref: '#/components/headers/x-fapi-interaction-id'
          content:
            application/json:
              schema:
                $ref:
                     '#/components/schemas/AccountPaymentNetworkList'
# ...
        '404':
# ...
        '500':
          $ref: './fdxapi.components.yaml#/components/responses/500'
        '501':
          $ref: './fdxapi.components.yaml#/components/responses/501'
        '503':
          $ref: './fdxapi.components.yaml#/components/responses/503'
```

1.3.6 Identifying needed but untagged elements

From the above `/accounts/{accountId}/payment-networks` path definition we can investigate the various optional and required elements and see that they are not tagged in the current FDX API releases V6.4 and V5.4.

Data Providers must identify which elements they need to implement to support the overall functionality of each operation. In this API definition, there are 13 not tagged elements though most will be needed:

```
# ...
# FapiInteractionIdHeader is required, and two optional parameters
    parameters:
      - $ref: '#/components/parameters/FapiInteractionIdHeader'
      - $ref: '#/components/parameters/FdxApiActorTypeHeader'
      - $ref: '#/components/parameters/FdxApiDataProviderIdHeader'

# ...
# AccountIdPath is required, OffsetQuery is deprecated
# and two other pagination parameters
    parameters:
        - $ref: '#/components/parameters/AccountIdPath'
        - $ref: '#/components/parameters/OffsetQuery'
        - $ref: '#/components/parameters/PageKeyQuery'
        - $ref: '#/components/parameters/LimitQuery'
      responses:
        '200':
# '200' response is not tagged but required for successful response
# of AccountPaymentNetworkList

          headers:
# Optional response header
            x-fapi-interaction-id:
              $ref: '#/components/headers/x-fapi-interaction-id'

# ...
# API-specific error response
        '404':
          # ...
# Three shared error responses
        '500':
          $ref: './fdxapi.components.yaml#/components/responses/500'
        '501':
          $ref: './fdxapi.components.yaml#/components/responses/501'
        '503':
          $ref: './fdxapi.components.yaml#/components/responses/503'
```

1.4 CSDF implementation using only tagged elements

A data provider planning to implement only the CSDF-defined APIs could create a FDX CSDF API subset by following these steps:

1. Identify whether you will be supporting both REG_E and REG_Z or just one of them
2. Identify the files which contain APIs you will implement, among fdxapi.core.yaml, fdxapi.customer.yaml, fdxapi.meta.yaml and fdxapi.money-movement.yaml, along with fdxapi.components.yaml
    1. fdxapi.extensions.yaml is used for documentation only, and not needed for an implementation
3. In those files, remove any API paths or methods which are not tagged (or have a tag not being implemented)
4. In those files, remove any API paths or methods which **are** tagged, but:
    1. Whose tag is not being implemented (i.e. just REG_E or REG_Z, per #1), OR
    2. References data not held by the data provider, OR
    3. References data which the data provider deems not to represent Covered Data
5. For the remaining API paths and methods, identify which of the untagged elements (as described above) they will choose or not choose to implement
6. Remove any untagged elements chosen not to implement
7. Remove any schemas remaining in the `components: schemas:` specification section which are not tagged or which have a tag that data provider is not implementing (unless it is referenced by an untagged element which IS being implemented)
8. Finally, evaluate all the tagged properties in the remaining tagged schemas and remove any:
    1. Whose tag is not being implemented (i.e. just REG_E or REG_Z, per #1), OR
    2. Is a data element not held by provider, OR
    3. Is a data element which provider deems not to represent Covered Data
9. These trimmed down files then represent the data provider's target API implementation for their CSDF

1.5 Related Resources

1. FDX Consensus Standard Data Format Guidelines document, published in both FDX API releases v6.4 and v5.4
2. Publication page for FDX API v6.4 - Spring 2025 Release
    1. V6.4 attached Reference List document
3. Publication page for FDX API v5.4
    1. V5.4 attached Reference List document
4. For background on the OpenAPI Specification and the use of custom extensions, see public sites:
    1. Official specification from openapis.org: https://spec.openapis.org/oas/latest.html#specification-extensions
    2. Brief guide from swagger.io: https://swagger.io/docs/specification/v3_0/openapi-extensions/
    3. FDX has officially registered our extension prefix '`x-fdx-`' at https://spec.openapis.org/registry/namespace/